SHAWN RAO

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EDUCATION

Bachelor of Science in Engineering (B.S.E.), Data Science *Expected 12/2027*
University of Michigan, College of Engineering Ann Arbor, MI

Relevant Coursework: SI 339 (Web Design, Development & Accessibility), SI 206 (Data-Oriented Programming), EECS 281 (Data Structures & Algorithms), EECS 203 (Discrete Math), EECS 376 (Foundations of CS), ROB 101 (Linear Algebra), MATH 215, STATS 250

In Progress (Fall 2026): DATASCI 306 (Statistical Computing in R), STATS 412 (Probability & Statistics), EECS 201 (Unix, Shell, Version Control)

Certificate, Applied Generative AI Specialization *06/2025 – 11/2025*
Purdue University Online Online

- Engineered retrieval-augmented generation pipelines integrating custom datasets with foundation models, improving response accuracy and reducing hallucination rate
- Built scalable AI agents in LangChain and Hugging Face, automating multi-step data analysis workflows end to end

TECHNICAL SKILLS

Languages: C++, Python, R, SQL, MATLAB, JavaScript (ES6+), HTML5, CSS3
Algorithms & Data Structures: Heaps and priority queues, union-find, graph algorithms (Prim, Dijkstra, MST), dynamic programming, branch and bound, hashing, BST/AVL trees, C++ STL
Systems & Shell: Linux, Unix, Bash, Zsh, shell scripting, process control (fork/exec/wait), sed, POSIX regex, SSH, WSL, Docker, AWS EC2/S3
Version Control & Testing: Git (interactive rebase, hooks, reflog recovery, conflict resolution, stash), GitHub, GitLab; randomized differential testing, mutation testing, Valgrind, strict-flag builds, WCAG audits
Data & Tools: Pandas, NumPy, tidyverse, SQL (window functions, recursive CTEs), LangChain, OpenAI API, Microsoft Excel, Word, PowerPoint
Front End: DOM manipulation, CSS Grid, Flexbox, media queries, React, Node.js

PROJECTS

Order-Matching Engine and Market Simulator *Fall 2026*
C++, STL Ann Arbor, MI

- Engineered a discrete-event matching engine pairing timestamped buy/sell orders against a price-time priority book, achieving $O(\log n)$ insert and match via paired binary heaps
- Produced running median price, per-trader totals and arbitrage reports in a single pass over each order stream, eliminating a second input read
- Diagnosed runtime bottlenecks on maximum-size inputs and replaced linear scans with a custom priority queue, selecting for asymptotic scalability over minimal code change

Shell Tooling and Git Automation *Fall 2026*
C, Bash, Zsh, Git, sed Ann Arbor, MI

- Constructed a Unix shell in C with fork, execvp and waitpid, handling cd and exit in parent scope since a child process cannot alter its parent's working directory
- Automated credential leak prevention via a pre-commit Git hook rejecting SSH private keys and hard-coded API credentials across all commits

EXPERIENCE

Computer Hardware Technician, Student Staff *04/2023 – 04/2026*
University of Michigan, College of Engineering (CAEN) Ann Arbor, MI

- Administered 200+ Linux, Windows and macOS lab workstations over 3 years, sustaining 99%+ uptime for 500+ students and faculty
- Established standardized imaging and diagnostics for student technicians, cutting hardware downtime 25% by ticket volume and issues resolved from updated runbooks
- Executed an enterprise CrowdStrike endpoint rollout to full coverage across a 200+ machine fleet, and integrated 3 new lab technologies, cutting setup time 40%
- Authored troubleshooting runbooks for incoming technicians maintaining a 200+ workstation fleet, eliminating repeat root-cause diagnosis of recurring failures

Information Technology Summer Intern *04/2021 – 07/2021*
NYX Inc. Livonia, MI

- Deployed employee workstations against company servers and reconciled IT hardware inventory across a live manufacturing site
- Resolved Windows and shop-floor issues for office and plant staff, prioritizing line uptime